THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · EASY LEVEL CHAPTER TEST SET 2

Chapter 11: **World of Algorithms** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The value of $65 \bmod 5$ (remainder on dividing 65 by 5) is: 1

(a) 0 **(b)** 1 **(c)** 2 **(d)** 13

2. The greatest of the common divisors of 14 and 30, found from their divisor lists, is: 1

(a) 1 **(b)** 4 **(c)** 2 **(d)** 3

3. The word ‘algorithm’ comes from the name of: 1

(a) Brahmagupta **(b)** Euclid **(c)** Al-Khwārizmī **(d)** Pythagoras

4. Set total = 0. For i = 1, 2, …, 14: if i is even, add i to total. The final value of total is: 1

(a) 58 **(b)** 57 **(c)** 55 **(d)** 56

5. The gcd algorithm stops when: 1

(a) the numbers are odd **(b)** both numbers are equal to 1 always
(c) one of the numbers becomes 0 **(d)** the sum is 10

6. Adding 84 and 23 by the column algorithm, the number of columns in which a carry is generated is: 1

(a) 0 **(b)** 1 **(c)** 2 **(d)** 3

7. Using the column addition algorithm, $321 + 299$ equals: 1

(a) 630 **(b)** 720 **(c)** 620 **(d)** 610

8. The divisor algorithm applied to 23 produces a list with how many entries? 1

(a) 2 **(b)** 1 **(c)** 3 **(d)** 4

9. The value of $90 \bmod 12$ (remainder on dividing 90 by 12) is: 1

(a) 8 **(b)** 5 **(c)** 7 **(d)** 6

10. The greatest of the common divisors of 28 and 20, found from their divisor lists, is: 1

(a) 3 **(b)** 5 **(c)** 4 **(d)** 8

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Run the divisor algorithm for 18: check j = 1, 2, …, 18. Write the list of divisors and say how many numbers are checked. 2

12. Find $58 \bmod 10$ and $68 \bmod 10$. What do you notice? 2

13. Add 366 and 260 using the column algorithm. State in which columns a carry is produced. 2

14. Run the divisor algorithm for 23: check j = 1, 2, …, 23. Write the list of divisors and say how many numbers are checked. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Find gcd(10, 25) by listing the divisors of both numbers. 3

16. Apply Euclid's subtraction algorithm to find gcd(45, 30). Write the sequence of pairs and count the subtractions. 3

17. Add 6680 and 5984 using the column algorithm. State in which columns a carry is produced. 3

OR

Find gcd(27, 21) by listing the divisors of both numbers. 3

18. Apply Euclid's subtraction algorithm to find gcd(45, 27). Write the sequence of pairs and count the subtractions. 3

OR

Add 5856 and 6761 using the column algorithm. State in which columns a carry is produced. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Trace this procedure: total = 0; for i = 1 to 6, if i is even add i to total. Write the values added and the final total. 5

OR

Find gcd(42, 48) by listing the divisors of both numbers. 5

20. Trace this procedure: total = 0; for i = 1 to 7, if i is even add i to total. Write the values added and the final total. 5

OR

Find gcd(44, 50) by listing the divisors of both numbers. 5

END OF QUESTION PAPER — Total: 40 Marks

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